DDTCDR: Deep Dual Transfer Cross Domain Recommendation

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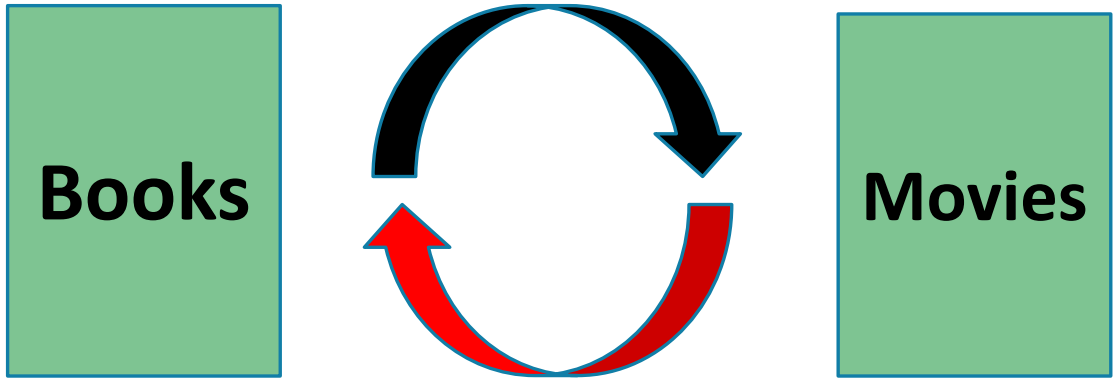
Main Idea of the Paper

Combines

- Dual Transfer Learning
- Latent Orthogonal Mapping between different domains

to provide cross-domain recommendations.

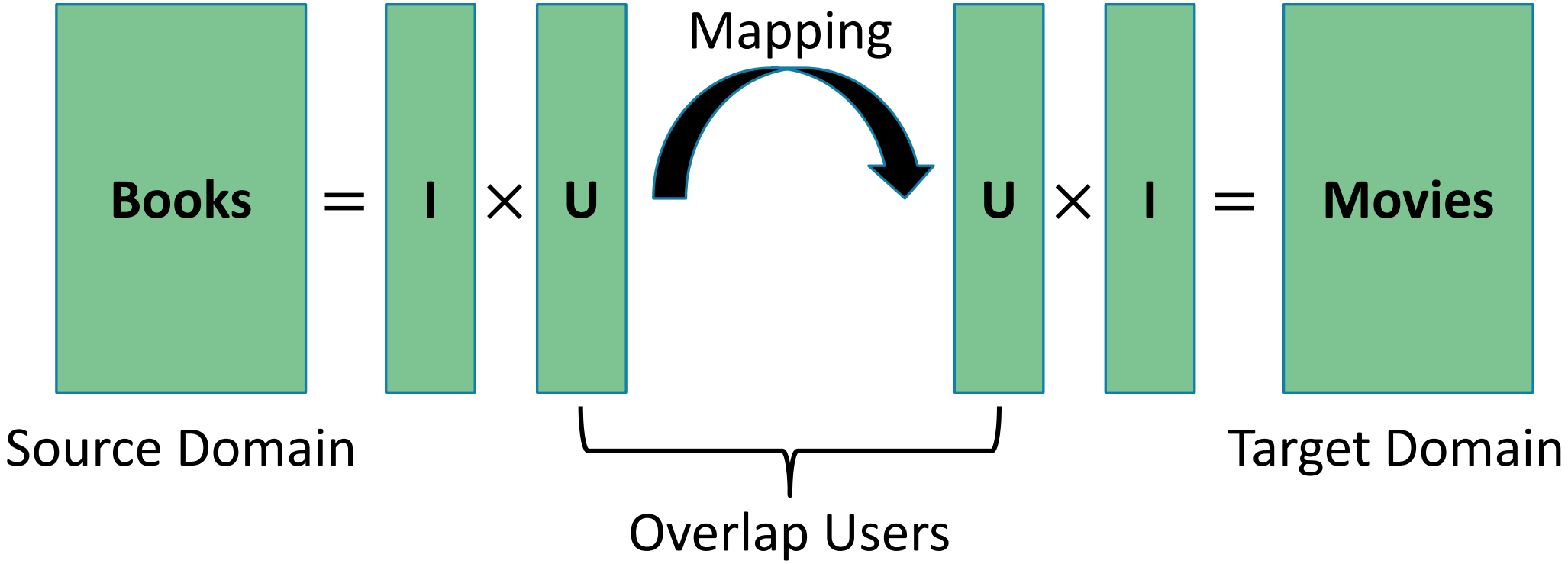


Latent Orthogonal Mapping

Cross Domain Recommendation (CDR)

- Assists recommendations in a target domain based on knowledge learned from a source domain. (Khan et al. 2017)
- Factorization-based approaches (Singh & Gordon 2008; Li et al. 2009; Hu et al. 2013; Loni et al. 2014; Sahebi et al. 2017; Wu et al. 2018) extract common features of users and items from both domains.
- Transfer Learning-based approaches (Lian et al. 2017; Hu et al. 2018) learn user preferences in the source domain and transfer them to the target domain.

Transfer Learning for CDR

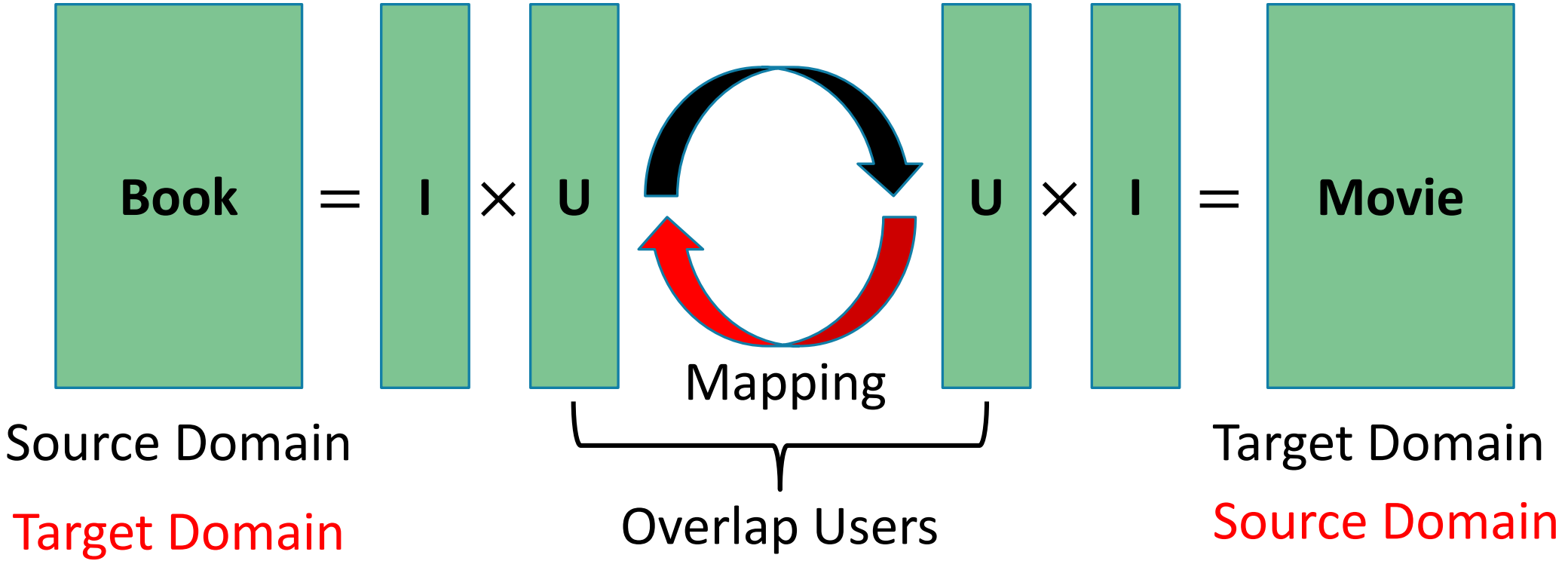


Dual Transfer Learning

- Bidirectionally transfers information between the source domain and the target domain. (Long et al. 2012)
- Successful applications:
  - Domain Adaption (Zhong et al. 2009)
  - Machine Translation (He et al. 2016; Xia et al. 2017; Wang et al. 2018).

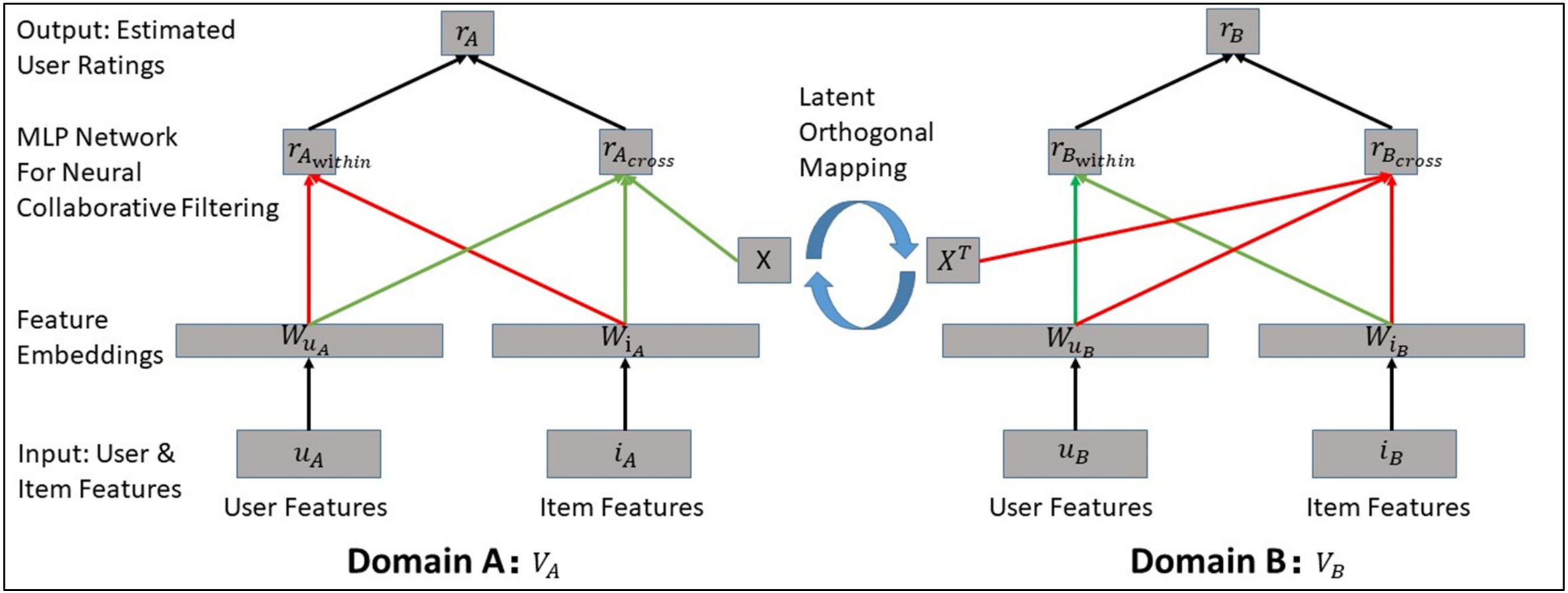
<u>Key Idea:</u> apply dual transfer learning to cross-domain recommendations!

Motivation: Dual Transfer Learning



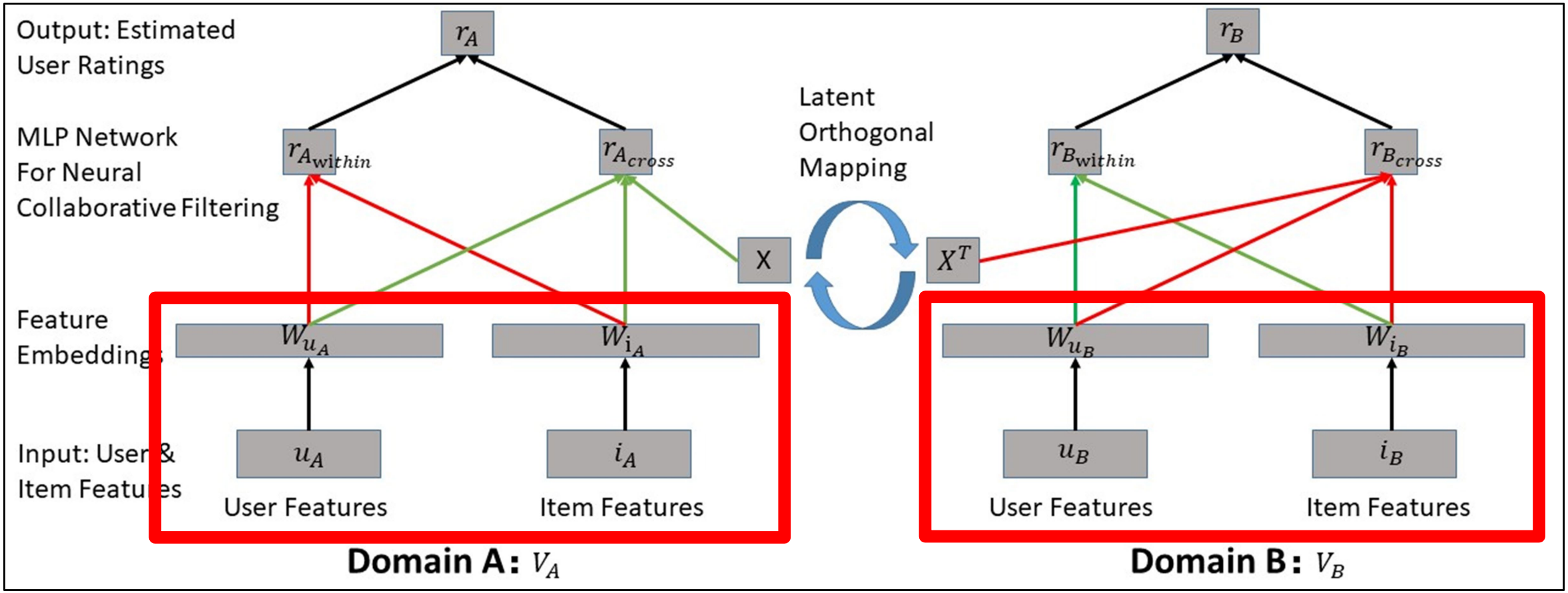
Improving recommendation performance in one domain would also lead to improvement in the other domain!

Model


Output: Estimated User Ratings
r_A
r_B
MLP Network For Neural Collaborative Filtering
$r_{A_{within}}$
$r_{A_{cross}}$
$r_{B_{within}}$
$r_{B_{cross}}$
Latent Orthogonal Mapping
X
X^T
Feature Embeddings
W_{u_A}
W_{i_A}
W_{u_B}
W_{i_B}
Input: User & Item Features
u_A
i_A
u_B
i_B
User Features
Item Features
User Features
Item Features
Domain A: V_A
Domain B: V_B

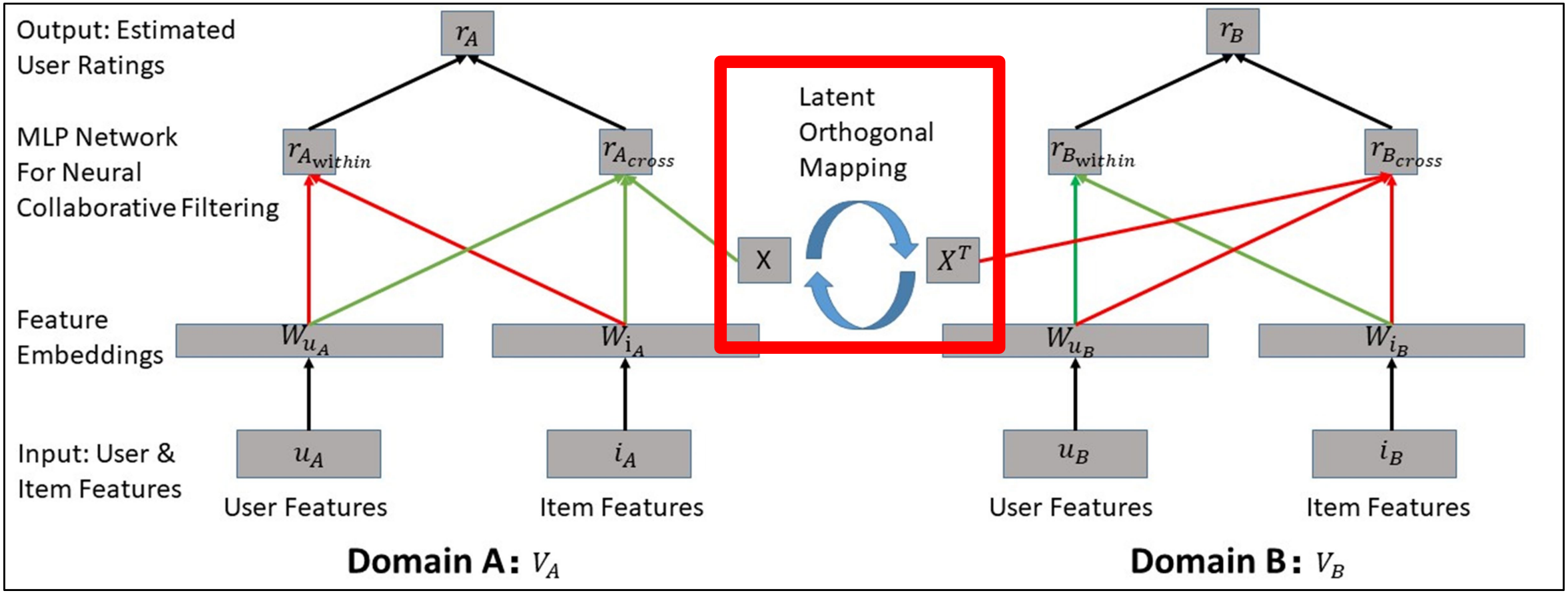
Model

Stage 1



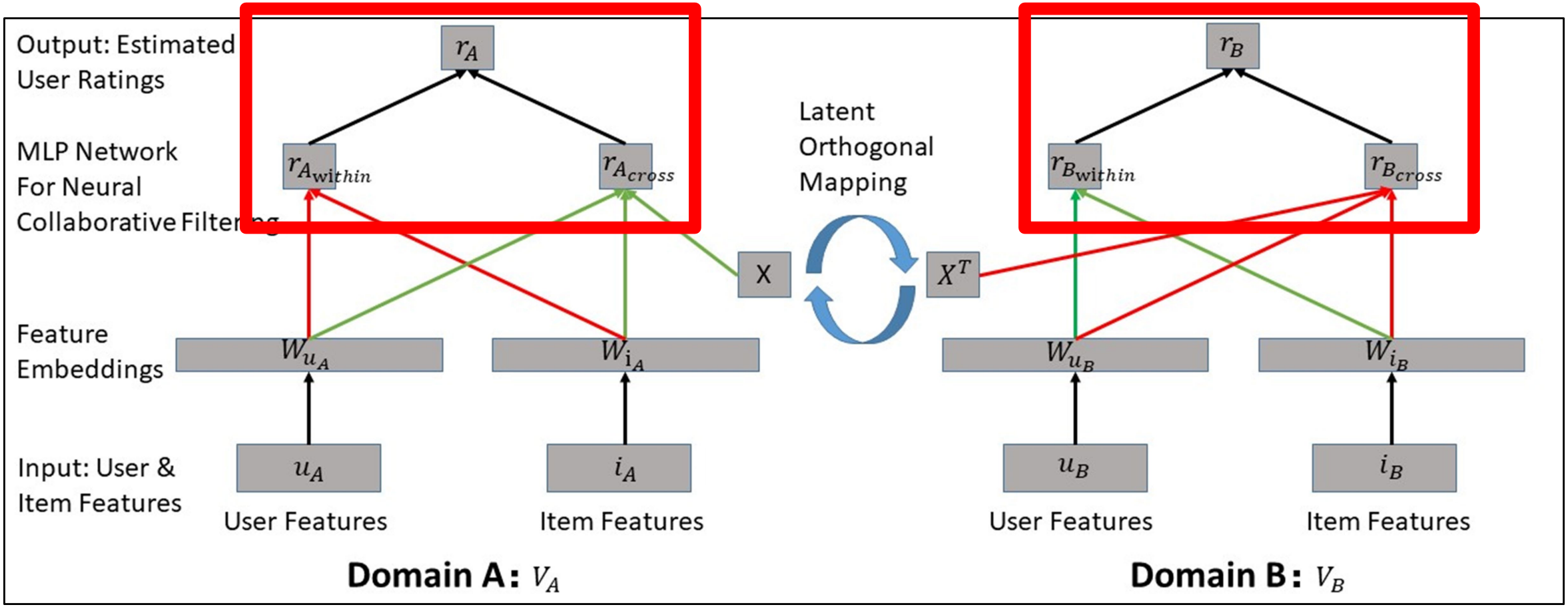
Model

Stage 2



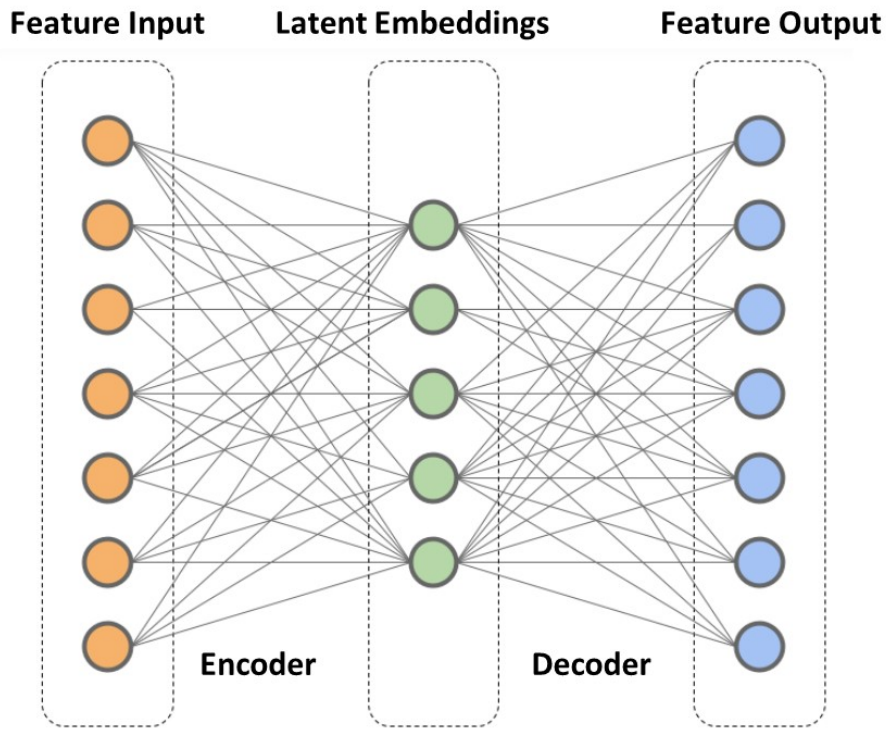
Model

Stage 3



Stage 1: Feature Autoencoding

- Maps user and item features into latent embeddings ($W_{u_a}, W_{u_b}, W_{i_a}, W_{i_b}$) through MLP network



Optimize reconstruction loss

$$L = ||u_a - MLP_{dec}(MLP_{enc}(u_a))||$$

$$W_{u_A} = MLP_{enc}(u_A)$$

$$W_{u_B} = MLP_{enc}(u_B)$$

$$W_{i_A} = MLP_{enc}(i_A)$$

$$W_{i_B} = MLP_{enc}(i_B)$$

Stage 2: Latent Orthogonal Mapping

- Learns the bidirectional orthogonal mapping (X, X^T) between user embeddings across different domains.

Orthogonality is important because it

- preserves similarities between user embeddings across different latent spaces.
- automatically derives the inverse mapping function.

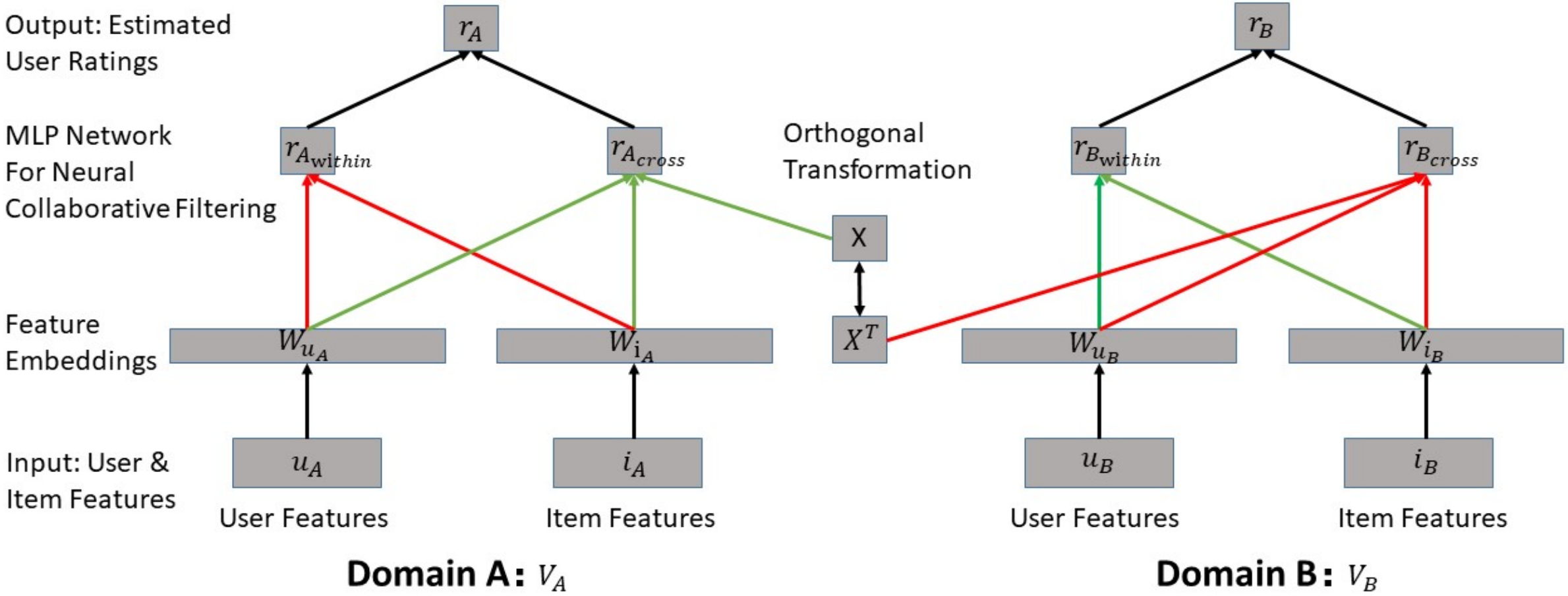
Stage 3: Deep Dual Transfer Learning

- Transfers user preferences from one domain to the other bidirectionally and provide rating estimation accordingly.
- We estimate ratings r using two components: within-domain user preference and cross-domain user preference
- Transfer Rate: α

$$r'_A = (1-\alpha)RS_A(W_{u_A}, W_{i_A}) + \alpha RS_B(X * W_{u_A}, W_{i_A})$$

$$r'_B = (1-\alpha)RS_B(W_{u_B}, W_{i_B}) + \alpha RS_A(X^T * W_{u_B}, W_{i_B})$$

Model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MLP Network For Neural Collaborative Filtering
Feature Embeddings
Input: User & Item Features
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Orthogonal Transformation
X
X^T
W_{u_A}
W_{i_A}
u_A
i_A
User Features
Item Features
Domain A: V_A
r_B
$r_{B_{within}}$
$r_{B_{cross}}$
W_{u_B}
W_{i_B}
u_B
i_B
User Features
Item Features
Domain B: V_B

Experiments: Data

Dataset: collected from an online recommendation service for books, movies, music

Contains rich information of user features and item features:

- User (Gender, Age, Movie Taste, Residence, Preference, Usage, Marital Status, Personality)
- Book (Category, Title, Author, Publisher, Language, Country, Price, Date)
- Movie (Genre, Title, Director, Writer, Runtime, Country, Rating, Votes)
- Music (Listener, PlayCount, Artist, Album, Tag, Release, Duration, Title)

Domain	Book	Movie	Music
# of Users	804,825	959,502	45,962
# of Items	182,653	79,866	183,114
# of Ratings	223,007,805	51,269,130	2,536,273
Sparsity	0.0157%	0.0669%	0.0301%

Experiments: Baselines

- Baseline Methods:
  - **CCCFNet**: Cross-domain Content-boosted Collaborative Filtering neural NETwork (Lian et al. 2017)
  - **CDFM**: Cross Domain Factorization Machine (Loni et al. 2014)
  - **CoNet**: Collaborative Cross Network (Hu et al. 2018)
  - **CMF**: Collective Matrix Factorization (Singh & Gordon, 2008)
  - **NCF**: Neural Collaborative Filtering (He et al. 2018)
- Conducted record-stratified 5-fold cross validation
- Evaluated performance using RMSE, MAE, Precision and Recall metrics

Results: Book/Movie Domains

Algorithm	Book				Movie			
	RMSE	MAE	Precision@5	Recall@5	RMSE	MAE	Precision@5	Recall@5
DDTCDR	**0.2213***	**0.1708***	**0.8595***	**0.9594***	**0.2213***	**0.1714***	**0.8925***	**0.9871***
Improved %	(+3.98%)	(+9.54%)	(+2.77%)	(+6.30%)	(+2.44%)	(+9.80%)	(+2.75%)	(+2.74%)
NCF	0.2315	0.1887	0.8357	0.8924	0.2276	0.1895	0.8644	0.9589
CCFNet	0.2639	0.1841	0.8102	0.8872	0.2476	0.1939	0.8545	0.9300
CDFM	0.2494	0.2165	0.7978	0.8610	0.2289	0.1901	0.8498	0.9312
CMF	0.2921	0.2478	0.7972	0.8523	0.2738	0.2293	0.8324	0.9012
CoNet	0.2305	0.1892	0.8328	0.8990	0.2298	0.1903	0.8680	0.9601

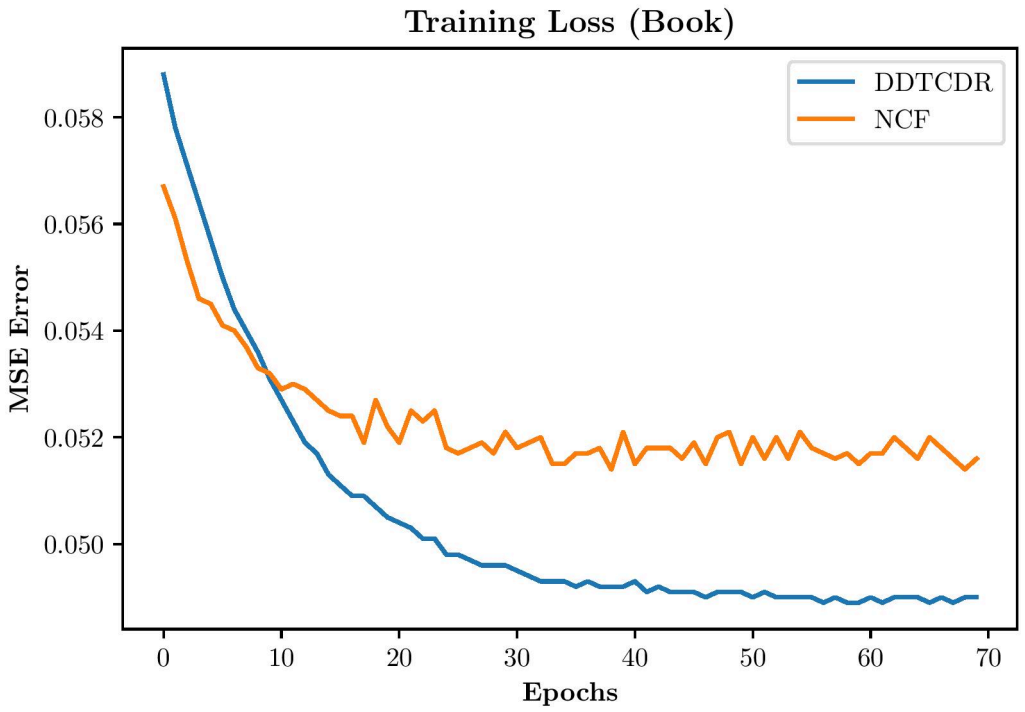
Results: Book/Music Domains

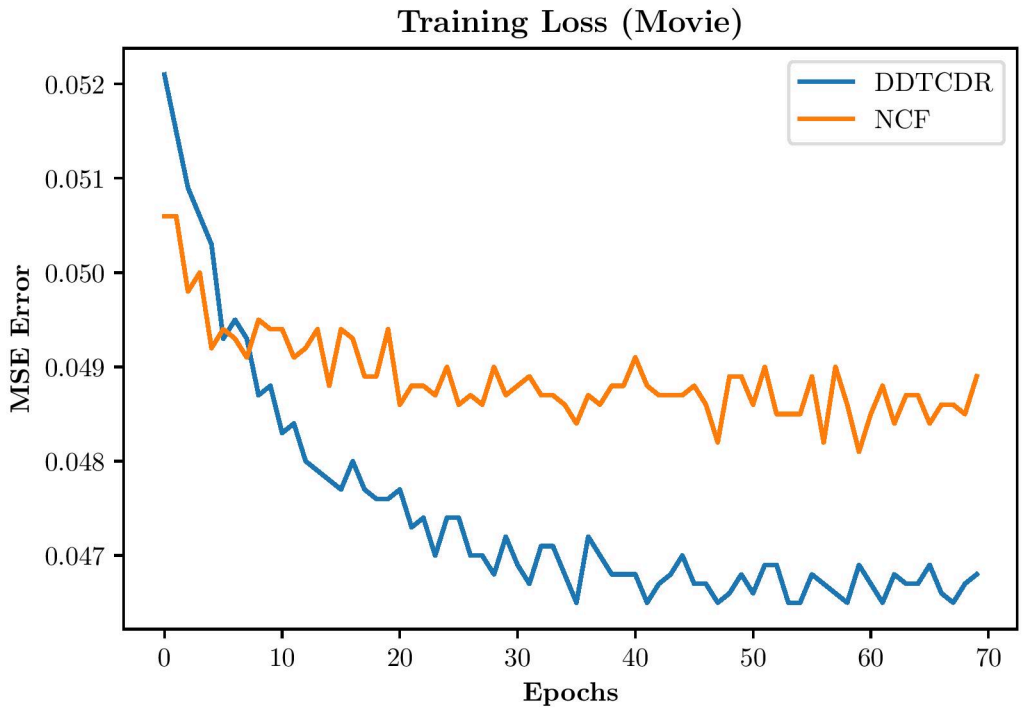
Algorithm	Book				Music			
	RMSE	MAE	Precision@5	Recall@5	RMSE	MAE	Precision@5	Recall@5
DDTCDR	**0.2209***	**0.1704***	**0.8570***	**0.9602***	**0.2753***	**0.2302***	**0.8392***	**0.8928***
Improved %	(+4.07%)	(+8.87%)	(+3.97%)	(+3.15%)	(+2.14%)	(+4.74%)	(+5.51%)	(+5.35%)
NCF	0.2315	0.1887	0.8230	0.9294	0.2828	0.2423	0.7930	0.8450
CCFNet	0.2630	0.1842	0.8150	0.9108	0.3090	0.2422	0.7902	0.8388
CDFM	0.2489	0.2155	0.8104	0.9102	0.3252	0.2463	0.7895	0.8365
CMF	0.2921	0.2478	0.8072	0.8978	0.3478	0.2698	0.7820	0.8324
CoNet	0.2307	0.1897	0.8230	0.9300	0.2801	0.2410	0.7912	0.8428

Results: Movie/Music Domains

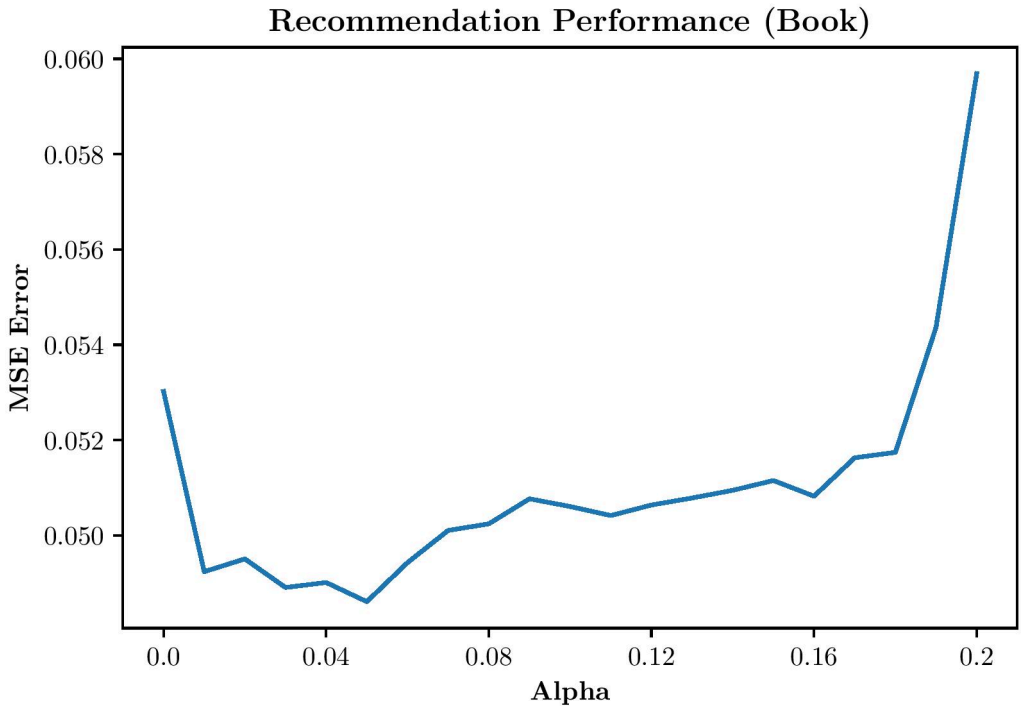
Algorithm	Movie				Music			
	RMSE	MAE	Precision@5	Recall@5	RMSE	MAE	Precision@5	Recall@5
DDTCDR	**0.2174***	**0.1720***	**0.8926***	**0.9869***	**0.2758***	**0.2311***	**0.8370***	**0.8902***
Improved %	(+3.75%)	(+9.77%)	(+5.32%)	(+3.68%)	(+1.89%)	(+4.24%)	(+4.30%)	(+4.38%)
NCF	0.2276	0.1895	0.8428	0.9495	0.2828	0.2423	0.7970	0.8501
CCFNet	0.2468	0.1932	0.8398	0.9310	0.3090	0.2433	0.7952	0.8498
CDFM	0.2289	0.1895	0.8306	0.9382	0.3252	0.2467	0.7880	0.8460
CMF	0.2738	0.2293	0.8278	0.9222	0.3478	0.2698	0.7796	0.8400
CoNet	0.2302	0.1908	0.8450	0.9508	0.2811	0.2428	0.8010	0.8512

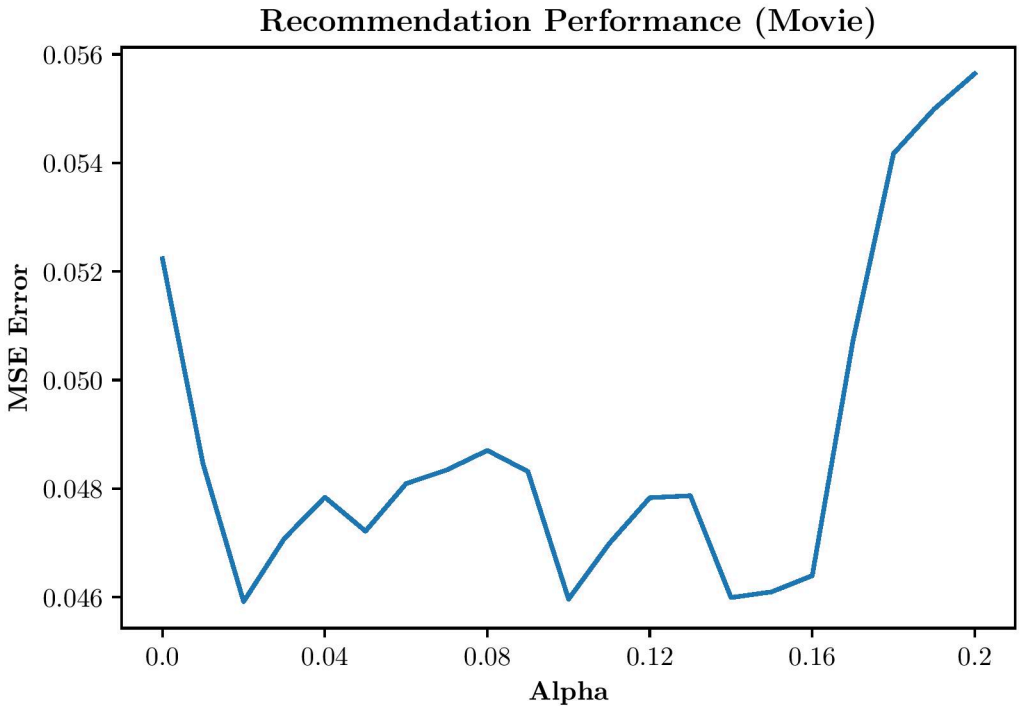
Results: Convergence





Results: Dual Transfer Rate α





Conclusions

- **Dual Transfer Learning significantly helps to improve cross-domain recommendations.**
- **Orthogonality of latent mapping plays an important role in dual transfer learning.**

Q&A

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